

UNITED STATES TOP 50 PLAYLIST

Performance and Song Popularity Trend Analysis

Research Paper • Data Analytics & Strategic Music-Market Intelligence

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Study period: 19 September 2026 – 28 September 2026

Dataset: 27,700 validated playlist observations across 554 daily snapshots

Scope: longevity, rank dynamics, artist visibility, popularity and content attributes

Executive Summary

This study analyzes historical United States Top 50 playlist snapshots to understand how songs and artists perform over time. Playlist position is treated as a repeated market-visibility signal, combined with popularity, longevity, artist presence, release format, explicitness, duration and album size.

The cleaned dataset contains 27,700 observations covering 18 May 2024 through 27 November 2025. It contains 943 unique songs and 297 unique artists, with no missing values in the analytical fields and playlist positions within the required 1–50 range.

The analysis moves beyond a simple current-rank view by evaluating sustained chart presence, average and best rank, popularity by rank tier, artist repeat appearances, release characteristics and content attributes. The observed Spearman relationship between playlist position and popularity is approximately -0.245, indicating a negative association in this dataset.

A Streamlit dashboard converts these results into an interactive stakeholder tool with date, artist, song, rank and album-type filters, plus timeline, song-trend, artist, popularity, content, release-format, duration and album-size views.

The project is descriptive and historical. It identifies patterns and decision-support signals; it does not claim that rank, duration, release format or explicitness causes popularity.

1. Introduction

1.1 Business Context

Streaming-era music markets generate repeated, time-stamped observations of what content remains visible. A daily Top 50 playlist therefore provides a longitudinal view of song and artist visibility rather than a single snapshot.

For music-market stakeholders, the analytical value is not limited to identifying the current top-ranked song. A stronger view considers persistence, movement, artist concentration, popularity and content attributes together.

1.2 Problem Statement

Despite access to daily Top 50 playlist observations, stakeholders may lack a structured way to understand ranking stability versus volatility, how long songs remain relevant, which artists repeatedly appear, and how song attributes relate to playlist performance.

1.3 Objectives

- Measure song longevity and sustained playlist presence.
- Characterize daily rank distribution and movement.
- Compare average rank, best rank and popularity across songs and rank tiers.
- Measure artist visibility through appearances and unique songs.
- Examine popularity versus playlist position.
- Compare explicit and non-explicit tracks.
- Compare album, single and compilation formats.
- Assess duration and album size in relation to popularity and rank.
- Translate the analysis into an interactive stakeholder dashboard.

2. Research Questions

- Which songs demonstrate the strongest sustained playlist presence?
- How do chart position and popularity move together?
- How does popularity differ across Top 10, Top 20 and Top 50 tiers?
- Which artists have the highest repeat playlist presence?
- How do explicit and non-explicit tracks differ in observed performance?
- How do album, single and compilation tracks compare?
- Is song duration associated with popularity or rank?
- Does album size show an observable relationship with song success?

3. Dataset and Data Quality

The project uses daily United States Top 50 playlist observations. Each observation represents a song's position and associated metadata on a specific playlist date.

Field	Description
date	Date of playlist snapshot
position	Playlist rank (1–50)
song	Song title
artist	Artist or credited artists
popularity	Popularity score
duration_ms	Song duration in milliseconds
album_type	Album / Single / Compilation
total_tracks	Number of tracks in the release
is_explicit	Explicit-content flag
album_cover_url	Album artwork URL

Validation of the cleaned analytical dataset showed 27,700 rows, 943 unique songs, 297 unique artists and 555 unique dates. Missing-value checks returned zero missing values, and positions were within the required 1–50 range.

4. Analytical Methodology

4.1 Data Preparation

The workflow loads the cleaned dataset, converts the date field to datetime, validates analytical columns, and prepares the data for longitudinal and categorical analysis.

4.2 Feature Engineering

Metric	Purpose
Days on Chart	Song longevity and sustained visibility
Average Rank	Overall playlist position
Best Rank Achieved	Strongest observed position
Rank Volatility Index	Stability versus movement
Popularity Trend	Historical popularity movement
Duration in Minutes	Interpretable track length
Rank Group	Top 10, Top 20 and Top 50 comparison

4.3 Analytical Approach

The analysis combines descriptive statistics, grouped aggregation, correlation analysis and interactive visualization. Spearman correlation is used for the monotonic relationship between playlist position and popularity. Findings are interpreted as observational associations.

5. Exploratory Data Analysis

5.1 Popularity and Rank

The observed Spearman correlation between playlist position and popularity is approximately -0.245. This indicates a modest inverse association: observations closer to the top of the playlist tend to have higher popularity scores.

Rank Group	Average Duration (min)	Median Duration (min)	Average Popularity
Top 10	3.32	3.11	89.61
Top 20	3.38	3.25	87.58
Top 50	3.28	3.13	87.02

The Top 10 has the highest average popularity among the three groups shown. Duration remains relatively close across the tiers.

5.2 Song Longevity

The longest observed chart presences include Something in the Orange (535 days), Last Night (491), I Remember Everything (445), See You Again (443), Stick Season (413), Cruel Summer (387), Lose Control (328), Thinkin’ Bout Me (323), Beautiful Things (306), and You Proof (301).

These results demonstrate why Days on Chart should be evaluated alongside peak position: sustained visibility represents a different performance signal from a brief peak.

5.3 Artist Visibility

The artist aggregation shows substantial repeat playlist presence. The highest observed appearance counts include Taylor Swift (2,014), Zach Bryan (1,854), Morgan Wallen (1,666), Sabrina Carpenter (966), Drake (838), and Chappell Roan (794).

Artist visibility is therefore evaluated through appearances, unique songs, average position and average popularity rather than appearance count alone.

5.4 Release and Content Attributes

The project compares album, single and compilation releases and evaluates explicit versus non-explicit tracks using observations, unique songs, average position and average popularity. Duration is converted from milliseconds to minutes, while total_tracks is used to examine album size.

6. Key Results and Insights

Longevity is a distinct performance dimension. Long-running tracks show that sustained playlist presence is not captured by peak rank alone.

Popularity and rank are related, but not identical. The negative correlation indicates a relationship while its moderate magnitude shows that popularity does not perfectly determine playlist position.

Artist presence is repeatable. A relatively small set of artists accumulates substantial playlist visibility across daily snapshots.

Top-tier tracks show higher observed popularity. Average popularity is 89.61 in the Top 10, compared with 87.58 in the Top 20 and 87.02 in the Top 50.

Duration is comparatively stable across rank tiers. Average and median duration remain close across rank groups, so duration should be evaluated with other variables.

Interactive exploration improves usability. The Streamlit dashboard lets stakeholders move from market-level patterns to specific artists, songs, dates and release formats.

7. Discussion

The results support a portfolio view of playlist performance. A robust assessment combines visibility, persistence, rank and popularity rather than relying on one metric. A track with a long chart life provides a different signal from a track that briefly reaches a very high position.

At artist level, breadth and intensity should also be separated. Playlist appearances measure visibility, unique songs show catalog breadth, average position describes typical placement, and popularity provides an additional engagement signal.

Because the dataset is observational, the project avoids causal claims. The appropriate interpretation is that the variables display measurable patterns or associations within the historical playlist data.

8. Recommendations

Use a visibility–persistence framework. Evaluate Days on Chart, Average Rank, Best Rank and Rank Volatility together.

Monitor artist portfolio presence. Track appearances and unique songs alongside average position and popularity.

Use rank tiers for benchmarking. Maintain Top 10, Top 20 and Top 50 benchmarks for consistent comparison.

Evaluate release format jointly with performance. Compare album, single and compilation tracks using rank, popularity, duration and time.

Treat content attributes as monitoring variables. Use explicitness, duration and album size descriptively and avoid causal conclusions.

Operationalize the dashboard. Use the Streamlit application for recurring filtered reviews without rebuilding the analysis.

Enrich future datasets. Add genre, release date, label, territory, marketing exposure and listener geography for deeper analysis.

9. Streamlit Dashboard

The interactive Streamlit application is the stakeholder-facing delivery layer of the project. It includes:

- Date range selector
- Artist and song filters
- Rank-range slider
- Album-type filter
- Executive KPI overview
- Playlist timeline and popularity trend
- Song ranking trend explorer
- Artist dominance leaderboard
- Popularity versus rank scatter plot
- Explicit versus non-explicit panel
- Release-format analysis
- Duration and album-size analysis
- Methodology documentation

10. Limitations and Future Work

- The dataset is observational and playlist-centric; associations should not be interpreted as causal effects.
- Popularity is an external score and is not a complete measure of listener engagement.
- Playlist curation, marketing exposure, advertising spend and algorithmic mechanisms are not directly observed.
- Listener geography, language, artist origin and campaign exposure are not in the supplied schema.
- Future versions could add genre, release dates, label, territory, campaign exposure and external streaming measures.
- Future work could formalize rank-volatility benchmarking and add statistical tests where feature definitions are consistently available.

11. Conclusion

This project demonstrates an end-to-end approach to historical music-market analytics. By combining daily playlist snapshots with engineered measures of longevity, rank, popularity, artist presence and content attributes, it provides a richer view of playlist performance than a single-day ranking.

The main contribution is the integration of song-level, artist-level and content-level perspectives into one decision-support framework, followed by an interactive dashboard that makes the analysis accessible to stakeholders.

The workflow is reproducible and extensible: future playlist observations can be added to the same pipeline, allowing the monitoring framework to evolve over time.

Appendix A — Core KPIs

KPI	Interpretation
Days on Chart	Longevity and sustained visibility
Average Rank	Overall playlist performance
Best Rank Achieved	Peak observed position
Rank Volatility Index	Stability versus movement
Popularity Score Trend	Historical popularity movement
Artist Dominance Index	Artist visibility / presence
Explicit Content Share	Content-mix monitoring

Appendix B — Project Deliverables

Deliverable	Purpose
01 — Data Ingestion & Validation	Analytical foundation
02 — Feature Engineering	Derived performance metrics
03 — EDA	Exploratory analysis and visualizations
04 — Insights	Key analytical findings
Streamlit Dashboard	Interactive stakeholder analytics
Research Paper	Business-facing analytical narrative
Executive Summary	Stakeholder communication

Dashboard Overview



The dashboard overview exposes the dataset scale and the main KPIs at first glance: 943 songs, 297 artists, average popularity of 87.65, and average chart position of 25.50.

Metric	Observed value
Records	27,700
Unique songs	943
Unique artists	297
Average popularity	87.65
Average chart position	25.50

Playlist Timeline & Popularity Trends

The timeline explorer shows how average playlist position and average popularity change across the historical period. The separate popularity trend makes longer-term changes easier to inspect.



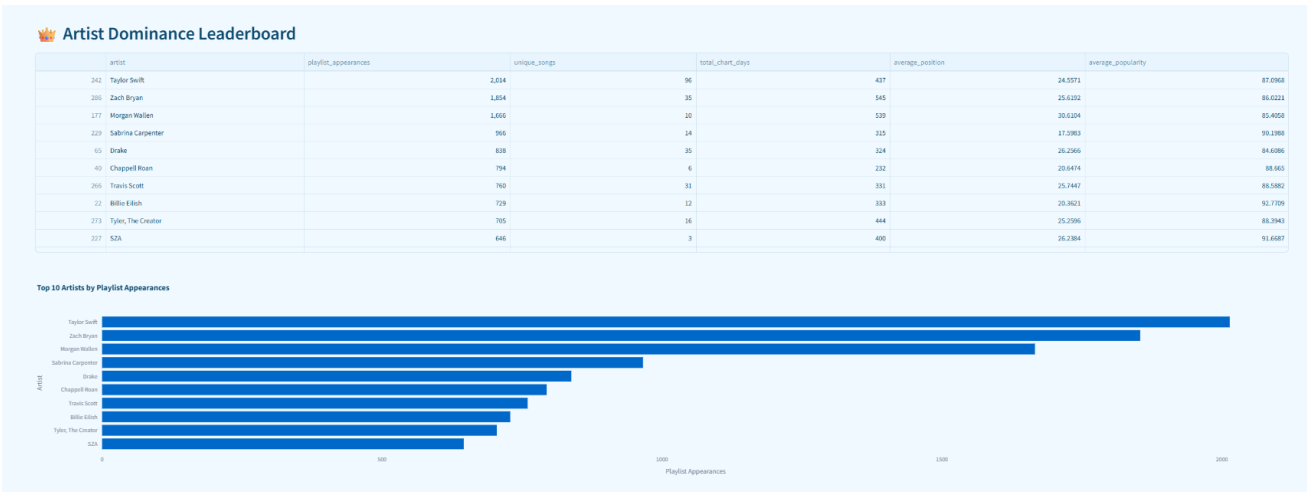
Song-Level Ranking Analysis

The song selector allows a specific track to be followed over time, making entry, movement, persistence, and decline visible rather than relying only on a single rank.



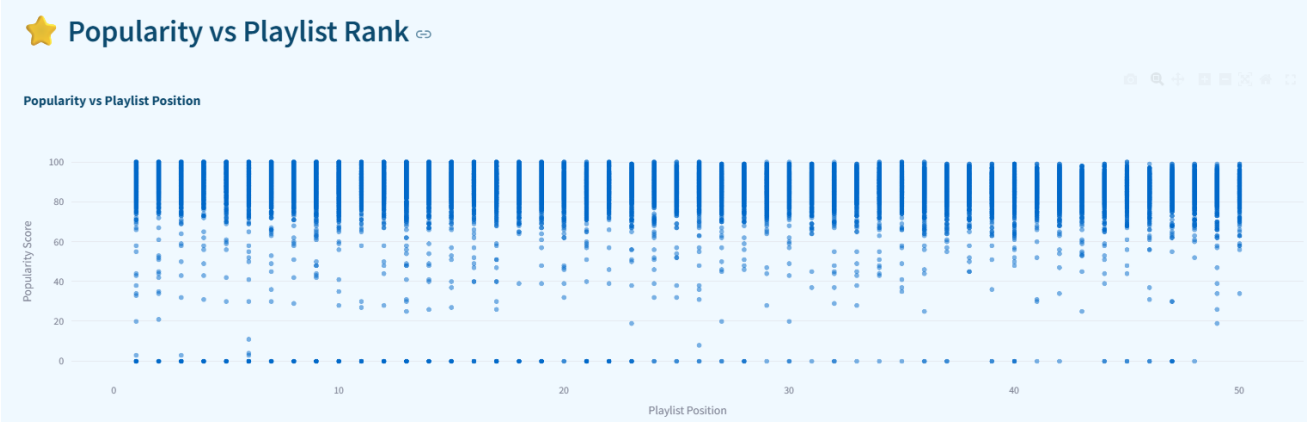
Artist Visibility & Dominance

The artist leaderboard combines playlist appearances, unique songs, chart days, average position, and average popularity. This supports analysis of repeat visibility rather than looking only at one peak position.



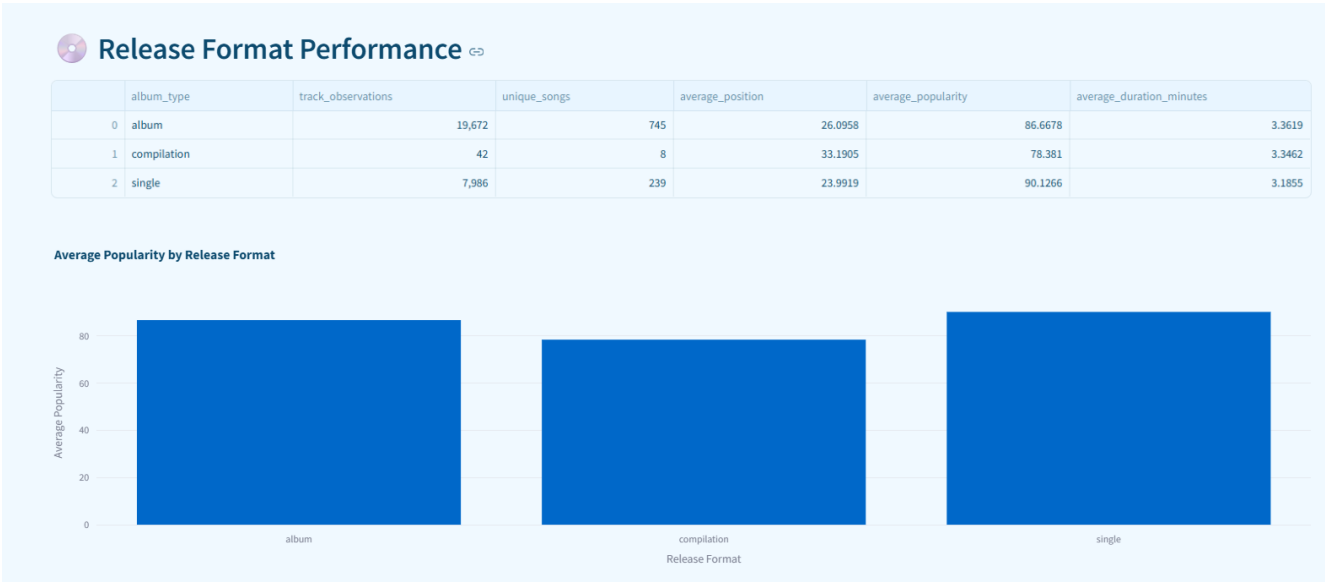
Popularity vs Playlist Rank

The scatter view compares popularity scores with playlist position. The dashboard also reports a Spearman rank-popularity correlation of -0.245 for the displayed dataset.



Content Attribute Analysis

The dashboard compares explicit and non-explicit observations and examines release format. These views address the project requirement to relate song attributes to playlist performance.

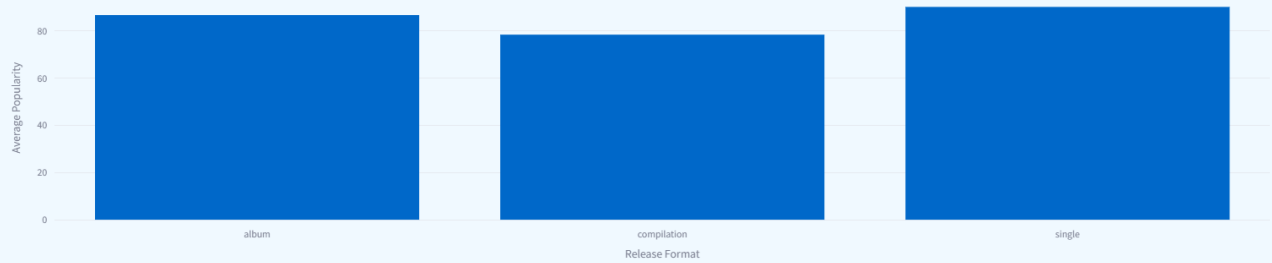




Release Format Performance [GO](#)

	album_type	track_observations	unique_songs	average_position	average_popularity	average_duration_minutes
0	album	19,672	745	26.0958	86.6678	3.3619
1	compilation	42	8	33.1905	78.381	3.3462
2	single	7,986	239	23.9919	90.1266	3.1855

Average Popularity by Release Format



Duration & Album Size Analysis

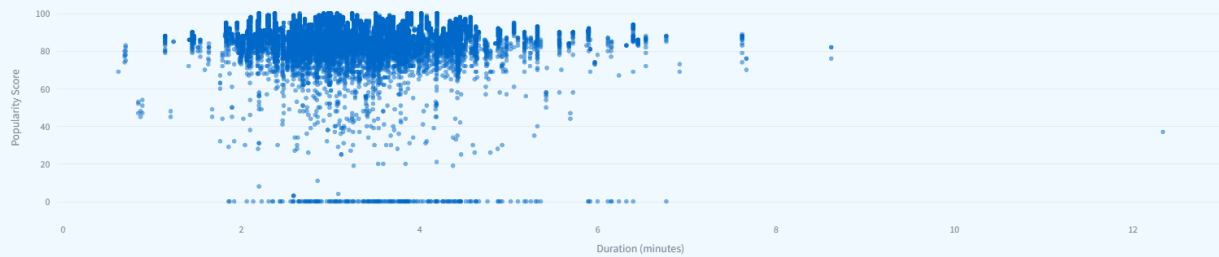
Duration analysis compares Top 10, Top 20, and Top 50 groups, while album-size analysis explores total tracks against average popularity and playlist position.

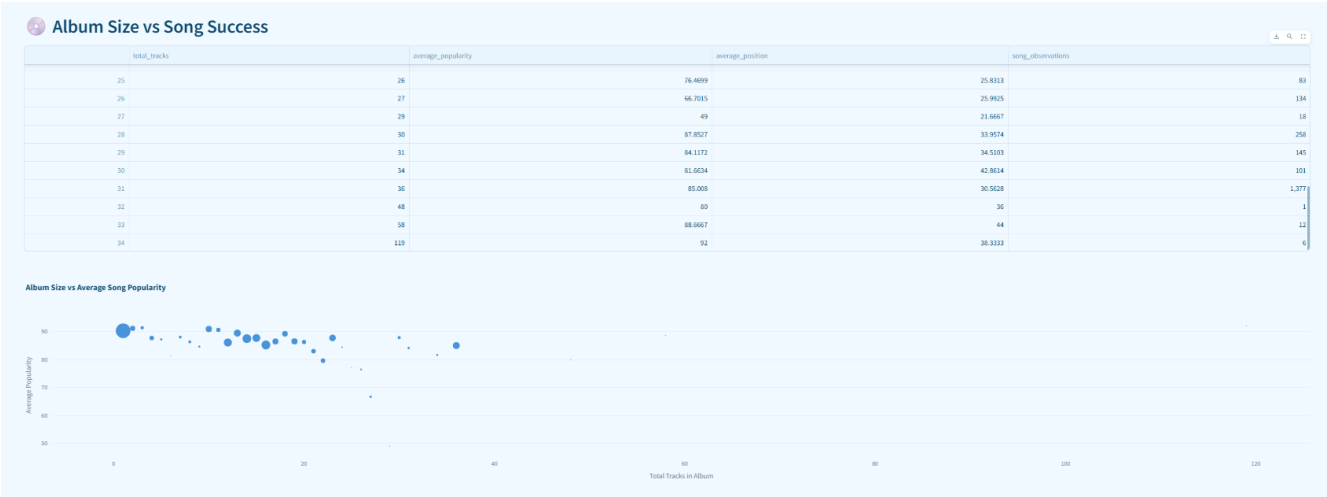


Song Duration Analysis

	rank_group	average_duration_minutes	median_duration_minutes	average_popularity
0	Top 10	3.3241	3.1061	89.6072
1	Top 20	3.3835	3.2487	87.5823
2	Top 50	3.2825	3.1287	87.0241

Song Duration vs Popularity





Methodology

The methodology section documents data coverage, chart position, popularity, chart days, average rank, artist presence, and duration. It also explicitly states that the dashboard focuses on historical analytics rather than prediction or recommendation.

Methodology

Data Coverage

The dashboard analyzes historical United States Top 50 playlist snapshots from May 2024 to November 2025.

Key Metrics

- Chart Position:** Daily playlist rank from 1 to 50.
- Popularity:** Popularity score associated with each track.
- Chart Days:** Number of days a song appears in the playlist.
- Average Rank:** Mean playlist position across observations.
- Artist Presence:** Number of playlist appearances and unique songs.
- Duration:** Track length converted from milliseconds to minutes.

Analysis Focus

The dashboard focuses on historical playlist performance, song popularity, artist visibility, ranking behavior, and content attributes. It does not perform prediction or recommendation.

This project demonstrates more than chart creation. It shows a complete analytics workflow: structured data preparation, derived business metrics, exploratory analysis, interactive filtering, dashboard design, and communication of findings.

What this demonstrates :

- Data skills: loading, cleaning, validation, date handling, aggregation and feature engineering.
- Analytical thinking: longevity, rank stability, artist presence, popularity relationships and content attributes.
- Visualization: time-series charts, ranking trends, leaderboards, scatter plots and comparison views.
- Product thinking: interactive filters let a stakeholder investigate the data instead of viewing a static report.
- Communication: the dashboard starts with KPIs and then moves from overall trends to song, artist and attribute analysis